

MATH 263 Notes
Chapter 4 - Elementary Number Theory and Methods
of Proofs

Natalie Piltoyan

March 2026

Contents

Direct Proof and Counterexample Introduction (4.1)	2
Proof-Writing Advice (4.2)	5
Definitions, Theorems, and all these terms	7
Rational Numbers (4.3)	8
Divisibility (4.4)	9
Division into Cases, Quotient Remainder Theorem (4.5)	11
Contradiction and Contrapositive (4.7)	13
Two Famous Theorems (4.8)	15

Direct Proof and Counterexample Introduction (4.1)

In this section, we will introduce some number theoretic definitions and give an introduction to writing proofs. Some of these terms may feel familiar, but you do need to practice using a definition to prove a statement.

Definition 1. Let $n \in \mathbb{Z}$. We say n is **even** if, and only if, there exists an integer k such that $n = 2k$.

Definition 2. Let $n \in \mathbb{Z}$. We say n is **odd** if, and only if, there exists an integer k such that $n = 2k + 1$.

For the definition of odd, you can also use $n = 2k - 1$. To prove that a number is even or odd, you need to find an integer k that satisfies that definition.

Examples

1. Is -8 even?

Solution Yes, since $-8 = 2 * (-4)$.

2. Is 7 even?

Solution No, since there is no integer n so that $7 = 2n$.

Definition 3. Let n be a positive integer ($n \in \mathbb{Z}^+$). We say n is **prime** if, and only if, $n > 1$ and for all positive integers a and b , $n = ab$ implies that $a = 1$ and $b = n$ or that $a = n$ and $b = 1$.

Definition 4. Let n be a positive integer ($n \in \mathbb{Z}^+$). We say n is **composite** if, and only if, $n > 1$ and there exist positive integers a and b such that $n = ab$ and

$$1 < a < n,$$

$$1 < b < n.$$

Do not be intimidated by long definitions. These definitions are just saying that a number is prime if its only factors are itself and 1, and that a number is composite if it has factors other than 1 and itself.

Generally, to prove an existential statement, you need to find one example that makes the statement true. For example, consider the statement

There exists a natural number n such that $n^2 > 5$,

To prove this is true, you only need to find one natural number whose square is larger than 5. Take $n = 3$: 3 is a natural number and $3^2 = 9 > 5$, so the statement is true.

In Chapter 3, we saw that the negation of a universal statement is an existential statement: the negation of $\forall x \in D, P(x)$ is $\exists x \in D$ such that $\neg P(x)$. So, to disprove a universal statement, you need to find an example that makes the negation true. This is a *counterexample*.

Sometimes, it takes a while to find the example or counterexample. If it doesn't feel obvious, that is ok! You will build that intuition up with practice.

To prove a universal statement, we sometimes can use the *method of exhaustion*, where you check that the statement holds for every single element in the domain. For example, consider the statement

For all integers n , if n is even and $4 \leq n \leq 20$, then n can be written as the sum of two prime numbers.

You could check that this statement holds for every even integer between 4 and 20 (4, 6, 8, 10, 12, 14, 16, 18, 20). However, this only works when our domain is finite, and it can be exhausting. This also does not work for domains such as $\mathbb{N}$ and $\mathbb{Z}$, which are both *countably infinite*.

The more common method is *universal generalization*. The idea here is that you pick an arbitrary element from your domain, show that the element has a particular property, and since the element was arbitrarily chosen, then every element in that domain has that property.

The **Method of Direct Proof** is a proof of a universal conditional statement that employs universal generalization. In other words, given a statement of the form

For all $x \in D$, if $P(x)$ then $Q(x)$,

A direct proof follows these steps:

1. Chooses a particular but arbitrary x from the domain D for which $P(x)$ is true.
2. Shows that this implies that $Q(x)$ must also be true using definitions, previously established results, and/or the rules of logical inference.

Below is an example of this method.

Theorem 1. *The sum of any two even numbers is even.*

We can rewrite this statement as

For all integers n, m , if n and m are even, then $n + m$ is even.

To prove this, we need to choose two arbitrary even integers, and show that their sum is even. When using the definition of even, make sure not to use the same variable for both. The proof goes as follows:

Proof. Suppose m and n are even integers. By the definition of even, there exist integers a and b such that

$$m = 2a \quad \text{and} \quad n = 2b.$$

Then $m + n = 2a + 2b$. Factoring out a 2, we have that

$$2a + 2b = 2(a + b).$$

Since a and b are integers, $a + b$ is an integer. Therefore, it follows that $m + n$ is an even integer. $\square$

Your proof may look slightly different and you may choose different wording. This is ok! However, your proof should start and end similarly - you should start with picking particular but arbitrary even integers, and end with showing that their sum is even.

From Chapter 3, we also know that the negation of an existential statement is a universal statement: the negation of $\exists x \in D, P(x)$ is $\forall x \in D, \neg P(x)$. So, to disprove an existential statement, it is the same as proving a universal statement is true.

Proof-Writing Advice (4.2)

A proof is an argument that you are making to a reader. Through your proof, you are trying to communicate something about a statement. There are many different ways to write a proof, and all are perfectly acceptable. However, there are some guidelines and rules that they all follow:

1. Start with *Proof* and end with $\square$. This signals where your proof starts and where it ends.
2. For universal quantified statements, you will almost always start with taking an arbitrary element of your domain that makes the hypothesis true. This should follow the format of “Suppose x is an arbitrary element of D ” or “Let x be an arbitrary element of D ”.
3. Your sentences should be grammatically correct. Shorthand notation is ok (such as $\forall x \in \mathbb{Z}$ instead of “for all integers x ”), but add punctuation where needed and use English when possible.
4. Make sure you explain what each variable is before using it.
5. Use “filler” words. Words like “then”, “therefore”, “so”, “because”, “thus” are helpful in understanding the flow of your proof.
6. If you reference a theorem or a definition, say that. If you are proving something about even numbers, you should write “by the definition of even, ...”. We will later see theorems that either have names or are numbered.
7. Explain your steps! The reader should not have to get a paper and pencil out to verify your algebraic steps.

Below are some common mistakes.

1. Using an example - If you are trying to prove something about a universal quantified statement, one example is not enough. You have to show that the statement holds for every single element of the domain.
2. Using the same letter to mean two things - Make sure to use different variables.
This often happens in proofs involving multiple numbers. Suppose you are trying to prove the statement from earlier, “For all integers a and b , if a and b are even, then $a + b$ is even.” If you were to write the following:

Let a and b be even integers. Since a is even, there exists an integer n such that $a = 2n$. Since b is even, there exists an integer n such that $b = 2n$.

The problem here is that you are indirectly assuming that a and b are equal.
Always use different letters for different variables. If you run out of English letters, there are other alphabets.

3. Assuming the conclusion is true before proving it - this is a fallacy.

Definitions, Theorems, and all these terms

As an aside, I thought it would be good to define what all these different terms are.

- Definition - an explanation of the meaning of a word.
- Theorem - a statement that has been proven to be true.
- Proposition - a true statement, often less interesting.
- Lemma - another true statement, also less interesting than a theorem
- Corollary - a true statement that is deduced from a theorem or proposition.
- Conjecture - a statement that is believed to be true, but has no formal proof (yet).
- Axiom - a statement we assume and accept to be true.

Many of these are similar. Don't worry too much about whether a statement is a proposition versus a lemma versus a corollary, the important fact is that they are all true and provable (minus a conjecture).

Rational Numbers (4.3)

Definition 5. Let r be a real number. We say r is **rational** if, and only if, there exist integers a and b with $b \neq 0$ such that $r = \frac{a}{b}$.

If a real number is not rational, we say it is **irrational**.

Examples

1. Is $\frac{2}{7}$ rational?

Solution Yes, because 2 and 7 are both integers and $7 \neq 0$.

2. Is 0.125 rational?

Solution Yes, because 0.125 can be rewritten as $\frac{125}{1000}$.

3. Is $\frac{2}{0}$ rational?

Solution No.

We use the symbol $\mathbb{Q}$ to denote the set of rationals and $\mathbb{R}$ to denote the set of real numbers. There are a lot of properties we can prove with rational numbers. Some of these may feel obvious and trivial, but the point is for you to learn how to prove statements using *only* the definitions.

The Zero Property. Let a and b be real numbers. If $ab = 0$, then $a = 0$ or $b = 0$. In other words, if the product is 0, then at least one of a and b must be 0.

The contrapositive also holds - given real numbers a and b , if $a \neq 0$ and $b \neq 0$, then $ab \neq 0$. For now, you can assume the Zero Property is true without proving it. We'll prove it in 4.5 with Division into Cases.

Theorem 2. Every integer is a rational number.

This follows since every integer n can be rewritten as $\frac{n}{1}$.

Theorem 3. The sum of any two rational numbers is rational.

Proof. Let p and q be rational numbers. Then there exists integers a, b, c, d with $b \neq 0$ and $d \neq 0$ such that

$$p = \frac{a}{b} \quad \text{and} \quad q = \frac{c}{d}.$$

Their sum is given by

$$p + q = \frac{ad + bc}{bd}.$$

Since a, b, c, d are all integers, $ad + bc$ and bd are both integers. Additionally, by the Zero Product Property, since $b \neq 0$ and $d \neq 0$, $bd \neq 0$. Thus, $p + q$ is rational. $\square$

Divisibility (4.4)

Definition 6. Let n and d be integers, $d \neq 0$. We say n is **divisible** by d if and only if there exists an integer k such that $dk = n$.

Here, we can say that d **divides** n . This is denoted by $d \mid n$.

If d does not divide n , we denote this with $d \nmid n$.

When we talk about divisibility, we *only* want integer factors. Yes, we can divide 19 by 4, but we will get a decimal (4.75), so we cannot say that 4 is a factor of 19.

Examples

1. Does $2 \mid 24$?

Solution Yes, $2 * 12 = 24$.

2. Does $3 \mid (6n + 15)$?

Solution Yes, because $6n + 3$ can be rewritten as $3(2n + 5)$ and this is divisible by 3.

3. Does $2 \mid 23$?

Solution No, since there is no integer k so that $2k = 23$.

Theorem 4. For all integers a and b , if a and b are positive and a divides b , then $a \leq b$.

This is saying that if both integers are positive, then no factor of a number can be greater than the number itself.

Theorem 5. For all integers a, b, c , if a divides b and b divides c , then a divides c .

This is the theorem on transitivity of divisibility, saying that the transitive property holds for divisibility as well.

Theorem 6. Any integer $n > 1$ is divisible by a prime number.

The proof for this goes as follows: you start with a positive integer n . If n is prime, then you are done. If not, then rewrite it as the product of two factors. If either of those factors are prime, then you are done; otherwise, rewrite each of those factors as the product of two factors. Keep doing this until you find one factor that is prime.

The following theorem was proposed by Euclid in ancient Greece (c. 300 BC) but proved by Gauss in 1801. It is also referred to as the Unique Factorization Theorem and as the Prime Factorization Theorem.

Theorem 7 (The Fundamental Theorem of Arithmetic). *For any integer $n > 1$, there exists a positive integer k , distinct prime numbers $p_1, p_2, \dots, p_k$ and positive integers $e_1, e_2, \dots, e_k$ such that*

$$n = p_1^{e_1} p_2^{e_2} \dots p_k^{e_k}.$$

This says that any positive integer can be rewritten as the product of prime numbers. This form is called **standard factored form**. Each p_k is a distinct prime number, and each e_k is an exponent. Below are a couple of examples.

Prime Factorization Examples

- $20 = 2^2 * 5$.
- $63 = 3^2 * 7$.
- $16016 = 2^4 * 7 * 11 * 13$.

Division into Cases, Quotient Remainder Theorem (4.5)

In the last section, if an integer d did not evenly divide into another integer n , then we said that $d \nmid n$. The Quotient Remainder Theorem shows us how to account for the remainder when we divide two integers.

Theorem 8 (Quotient Remainder Theorem). *Given any integer n and any positive integer d , there exist unique integers q and r such that $0 \leq r < d$ and*

$$n = qd + r.$$

Example

1. For $n = 56$, $d = 5$, find integers q and r so that $n = qd + r$ and $0 \leq r < d$.

Solution Let $q = 11$, $r = 1$. Then $56 = 11 * 5 + 1$.

If we want to compute just q or just r , we can use the following functions.

Definition 7. *Let n be an integer and d be a positive integer. Then $n \mathbf{div} d$ gives the integer quotient when n is divided by d , and $n \mathbf{mod} d$ gives the remainder when n is divided by d .*

In the quotient remainder theorem, $n \mathbf{div} d = q$ and $n \mathbf{mod} d = r$.

Example

1. Let $n = 56$, $d = 5$. Then $56 \mathbf{div} 5 = 11$ and $56 \mathbf{mod} 5 = 1$.
2. You order a package on a Monday. The company says it will take 32 days for it to arrive. On what day of the week will it arrive?

Solution $32 \mathbf{mod} 7 = 4$, so it will arrive 4 days after Monday, or Thursday.

3. This year, your birthday falls on a Thursday. Assuming that next year is not a leap year, what day of the week will your birthday be on next year?

Solution $365 \mathbf{mod} 7 = 1$, so your birthday will be 1 day after Thursday, or Friday.

An important property of the integers is that every integer is either even or odd. Sometimes, this is called *parity* (5 has odd parity, 6 has even parity). From this, we get the following theorem:

Theorem 9. *Any two consecutive integers have opposite parity.*

To prove this, we start with an integer n and the consecutive integer, $n + 1$. We still use a direct proof, but we have to consider two cases: when n is even and when n is odd.

In general, this proof technique is called **Division into Cases**. If there are multiple conditions in the premise of a statement, you need to show that the conclusion is true for each one of those conditions. You can think of this as an “if else if” algorithm.

Contradiction and Contrapositive (4.7)

Before we begin this chapter, we should revisit some material from Chapter 2.

- Every statement is either true or false, but it cannot be both. If the negation of a statement is false, then the original statement must be true.
- A *contradiction* is a statement form that is always false regardless of the truth values substituted in.
- Given a conditional statement “If p , then q ”, the *contrapositive* of this statement is “If not q , then not p ”. The contrapositive of a statement is logically equivalent to the original statement.

Sometimes, it is easier to prove that the negation of a statement is false rather than prove that the statement itself is true. This is called a **Proof by Contradiction**. Below are the steps:

1. Suppose the statement to be proved is false. So, suppose the negation of the statement is true.
2. Show that this leads to a contradiction.
3. Conclude that since the negation leads to a contradiction, the original statement must be true.

What that contradiction is varies by proof. In some proofs, you will contradict the hypothesis. In others, you will obtain something absurd (such as $1 = 0$ or $\frac{1}{2}$ is rational). In both cases, you conclude that the negation is false, so the original statement must be true. Proof by contradiction is used quite often in mathematics. Both of the main theorems in Section 4.8 use a proof by contradiction. Below is an example of a proof via contradiction.

Theorem 10. *There is no integer that is both even and odd.*

Proof. Suppose, by way of contradiction, the statement is false. That is, suppose there exists an integer n that is both even and odd. By the definition of even, there exists an integer a so that

$$n = 2a.$$

By the definition of odd, there exists another integer b so that

$$n = 2b + 1.$$

Then we have that

$$2a = 2b + 1.$$

Rearranging this, we see that

$$2a - 2b = 1$$

$$2(a - b) = 1$$

$$a - b = \frac{1}{2}.$$

Since a and b are integers, $a - b$ should be an integer. However, $\frac{1}{2}$ is not an integer, so we have reached a contradiction. Thus, the statement must be true, there is no integer that is both even and odd. $\square$

Another proof technique is **Proof by Contraposition**. The steps are shown below.

1. Write out the contrapositive of the statement.
2. Prove the contrapositive is true via a direct proof.

Since the contrapositive of a statement is logically equivalent to the statement, it follows that if the contrapositive is true, then the statement must also be true.

Both of the above proof techniques are called *indirect proofs*.

Two Famous Theorems (4.8)

We have clear definitions of most number groups - the natural numbers, integers, and rationals. However, the irrationals are defined as all real numbers that are not rational. So how do we know a real number is irrational?

Theorem 11. $\sqrt{2}$ is irrational.

The proof for this starts with assuming that the statement is false (by way of contradiction!), and assumes that $\sqrt{2}$ is rational. In doing this, we also assume that the fraction it is written as is fully simplified - the numerator and denominator have no common factors. This is the part that gets contradicted later on in the proof.

The following theorem was proven by Euclid in his work *Elements*. He actually wrote a direct proof for this, but the most common method used to prove it is also via contradiction.

Theorem 12. *There are infinitely many prime numbers.*